



Configuring Static Routes to Set Up a Small-Scale Network

Difficulty (out of 5 stars): ★★★

Recommended study time: 2 hours

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1 About This Lab

1.1 Task Description

Static routes are manually configured and maintained by network administrators. Static routes are easy to configure and widely used on networks. In addition, static routes can implement load balancing and route backup. Therefore, it is essential to learn and understand the application and configuration of static routes.

After completing this task, you will have a clear understanding of static routes and be able to independently use static routes to build and configure a small network.

1.2 Objectives

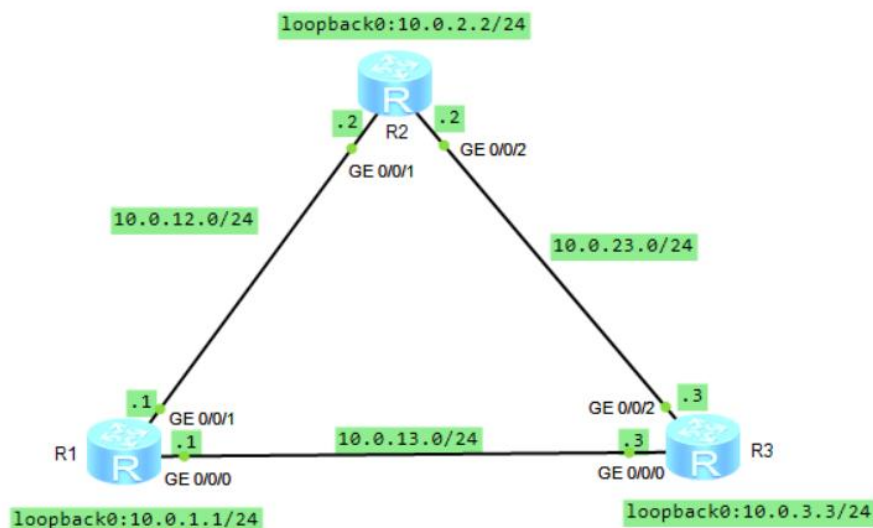
On completing this task, you will be able to:

1. Understand how to configure static routes.
2. Test static route connectivity.
3. Configure a default route to implement access between the local network and external network.
4. Configure backup static routes.

2 Task Preparation

2.1 Network Topology

Figure 2-1 Lab topology





2.2 Initial Configuration

- Initial configuration of R1

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R1
[R1]int lo0
[R1-LoopBack0]ip address 10.0.1.1 24
[R1-LoopBack0]quit
[R1]int g0/0/0
[R1-GigabitEthernet0/0/0]ip address 10.0.13.1 24
[R1-GigabitEthernet0/0/0]quit
[R1]int g0/0/1
[R1-GigabitEthernet0/0/1]ip address 10.0.12.1 24
[R1-GigabitEthernet0/0/1]quit
[R1]
```

- Initial configuration of R2

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R2
[R2]int lo0
[R2-LoopBack0]ip address 10.0.2.2 24
[R2-LoopBack0]quit
[R2]int g0/0/1
[R2-GigabitEthernet0/0/1]ip address 10.0.12.2 24
[R2-GigabitEthernet0/0/1]quit
[R2]int g0/0/2
[R2-GigabitEthernet0/0/2]ip address 10.0.23.2 24
[R2-GigabitEthernet0/0/2]quit
[R2]
```

- Initial configuration of R3

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R3
[R3]int lo0
[R3-LoopBack0]ip address 10.0.3.3 24
[R3-LoopBack0]quit
[R3]int g0/0/0
[R3-GigabitEthernet0/0/0]ip address 10.0.13.3 24
[R3-GigabitEthernet0/0/0]quit
[R3]int g0/0/2
[R3-GigabitEthernet0/0/2]ip address 10.0.23.3 24
[R3-GigabitEthernet0/0/2]quit
[R3]
```



3 Task Implementation

3.1 Checking Basic Configurations

Run the **display ip interface brief** command on R1, R2, and R3 to check interface configurations.

```
<R1>display ip interface brief
```

```
*down: administratively down
```

```
^down: standby
```

```
(l): loopback
```

```
(s): spoofing
```

```
The number of interface that is UP in Physical is 4
```

```
The number of interface that is DOWN in Physical is 1
```

```
The number of interface that is UP in Protocol is 4
```

```
The number of interface that is DOWN in Protocol is 1
```

Interface	IP Address/Mask	Physical	Protocol
GigabitEthernet0/0/0	10.0.13.1/24	up	up
GigabitEthernet0/0/1	10.0.12.1/24	up	up
GigabitEthernet0/0/2	unassigned	down	down
LoopBack0	10.0.1.1/24	up	up(s)
NULL0	unassigned	up	up(s)

```
<R1>
```

```
<R2>display ip int brief
```

```
*down: administratively down
```

```
^down: standby
```

```
(l): loopback
```

```
(s): spoofing
```

```
The number of interface that is UP in Physical is 4
```

```
The number of interface that is DOWN in Physical is 1
```

```
The number of interface that is UP in Protocol is 4
```

```
The number of interface that is DOWN in Protocol is 1
```

Interface	IP Address/Mask	Physical	Protocol
GigabitEthernet0/0/0	unassigned	down	down
GigabitEthernet0/0/1	10.0.12.2/24	up	up
GigabitEthernet0/0/2	10.0.23.2/24	up	up
LoopBack0	10.0.2.2/24	up	up(s)
NULL0	unassigned	up	up(s)

```
<R2>
```

```
<R3>display ip int brief
```

```
*down: administratively down
```

```
^down: standby
```

```
(l): loopback
```

```
(s): spoofing
```

```
The number of interface that is UP in Physical is 4
```

```
The number of interface that is DOWN in Physical is 1
```

```
The number of interface that is UP in Protocol is 4
```



The number of interface that is DOWN in Protocol is 1

Interface	IP Address/Mask	Physical	Protocol
GigabitEthernet0/0/0	10.0.13.3/24	up	up
GigabitEthernet0/0/1	unassigned	down	down
GigabitEthernet0/0/2	10.0.23.3/24	up	up
LoopBack0	10.0.3.3/24	up	up(s)
NULL0	unassigned	up	up(s)

<R3>

Run the **ping** command to check the connectivity between R1 and other devices.

```
<R1>ping 10.0.12.2
PING 10.0.12.2: 56 data bytes, press CTRL_C to break
  Reply from 10.0.12.2: bytes=56 Sequence=1 ttl=255 time=40 ms
  Reply from 10.0.12.2: bytes=56 Sequence=2 ttl=255 time=20 ms
  Reply from 10.0.12.2: bytes=56 Sequence=3 ttl=255 time=10 ms
  Reply from 10.0.12.2: bytes=56 Sequence=4 ttl=255 time=10 ms
  Reply from 10.0.12.2: bytes=56 Sequence=5 ttl=255 time=20 ms

--- 10.0.12.2 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 10/20/40 ms

<R1>ping 10.0.13.3
PING 10.0.13.3: 56 data bytes, press CTRL_C to break
  Reply from 10.0.13.3: bytes=56 Sequence=1 ttl=255 time=40 ms
  Reply from 10.0.13.3: bytes=56 Sequence=2 ttl=255 time=30 ms
  Reply from 10.0.13.3: bytes=56 Sequence=3 ttl=255 time=30 ms
  Reply from 10.0.13.3: bytes=56 Sequence=4 ttl=255 time=20 ms
  Reply from 10.0.13.3: bytes=56 Sequence=5 ttl=255 time=30 ms

--- 10.0.13.3 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 20/30/40 ms

<R1>
```

Run the **ping** command to check the connectivity between R2 and other devices.

```
<R2>ping 10.0.23.3
PING 10.0.23.3: 56 data bytes, press CTRL_C to break
  Reply from 10.0.23.3: bytes=56 Sequence=1 ttl=255 time=20 ms
  Reply from 10.0.23.3: bytes=56 Sequence=2 ttl=255 time=30 ms
  Reply from 10.0.23.3: bytes=56 Sequence=3 ttl=255 time=20 ms
  Reply from 10.0.23.3: bytes=56 Sequence=4 ttl=255 time=30 ms
  Reply from 10.0.23.3: bytes=56 Sequence=5 ttl=255 time=10 ms

--- 10.0.23.3 ping statistics ---
```



```
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 10/22/30 ms
```

```
<R2>
```

3.2 Testing the Connectivity Between R2 and the Destination Networks 10.0.13.0/24 and 10.0.3.0/24

```
<R2>ping 10.0.13.3
PING 10.0.13.3: 56 data bytes, press CTRL_C to break
Request time out
Request time out
Request time out
Request time out
Request time out
```

```
--- 10.0.13.3 ping statistics ---
5 packet(s) transmitted
0 packet(s) received
100.00% packet loss
```

```
<R2>ping 10.0.3.3
PING 10.0.3.3: 56 data bytes, press CTRL_C to break
Request time out
Request time out
Request time out
Request time out
Request time out
```

```
--- 10.0.3.3 ping statistics ---
5 packet(s) transmitted
0 packet(s) received
100.00% packet loss
```

```
<R2>
```

If R2 needs to communicate with the network segment 10.0.3.0/24, R2 must have a route to this network segment, and R3 must have a route to the IP network segment where the corresponding interface of R2 resides. The preceding check results show that R2 cannot communicate with 10.0.3.3 or 10.0.13.3.

Run the **display ip routing-table** command to check the routing table on R2. You can find that the routing table does not contain routes to the two network segments.

```
<R2>display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
      Destinations : 13          Routes : 13

Destination/Mask    Proto    Pre  Cost   Flags  NextHop    Interface
```



```

10.0.2.0/24 Direct 0 0 D 10.0.2.2 LoopBack0
10.0.2.2/32 Direct 0 0 D 127.0.0.1 LoopBack0
10.0.2.255/32 Direct 0 0 D 127.0.0.1 LoopBack0
10.0.12.0/24 Direct 0 0 D 10.0.12.2 GigabitEthernet 0/0/1
10.0.12.2/32 Direct 0 0 D 127.0.0.1 GigabitEthernet 0/0/1
10.0.12.255/32 Direct 0 0 D 127.0.0.1 GigabitEthernet 0/0/1
10.0.23.0/24 Direct 0 0 D 10.0.23.2 GigabitEthernet 0/0/2
10.0.23.2/32 Direct 0 0 D 127.0.0.1 GigabitEthernet 0/0/2
10.0.23.255/32 Direct 0 0 D 127.0.0.1 GigabitEthernet 0/0/2
127.0.0.0/8 Direct 0 0 D 127.0.0.1 InLoopBack0
127.0.0.1/32 Direct 0 0 D 127.0.0.1 InLoopBack0
127.255.255.255/32 Direct 0 0 D 127.0.0.1 InLoopBack0
255.255.255.255/32 Direct 0 0 D 127.0.0.1 InLoopBack0
<R2>

```

3.3 Configuring Static Routes on R2

Configure static routes to 10.0.13.0/24 and 10.0.3.0/24, and set the next hop to 10.0.23.3 (IP address of R3's G0/0/0). Static routes have a default preference value of 60, so you do not need to set their preferences manually.

```
[R2]ip route-static 10.0.13.0 24 10.0.23.3
```

```
[R2]ip route-static 10.0.3.0 24 10.0.23.3
```

Note: In the **ip route-static** command, 24 indicates the subnet mask length, which can also be written in the complete mask format 255.255.255.0.

```
[R2]display ip routing-table
```

Route Flags: R - relay, D - download to fib

Routing Tables: Public

Destinations : 15

Routes : 15

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.2.0/24	Direct	0	0	D	10.0.2.2	LoopBack0
10.0.2.2/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.2.255/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.3.0/24	Static	60	0	RD	10.0.23.3	GigabitEthernet 0/0/2
10.0.12.0/24	Direct	0	0	D	10.0.12.2	GigabitEthernet 0/0/1
10.0.12.2/32	Direct	0	0	D	127.0.0.1	GigabitEthernet 0/0/1
10.0.12.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet 0/0/1
10.0.13.0/24	Static	60	0	RD	10.0.23.3	GigabitEthernet 0/0/2
10.0.23.0/24	Direct	0	0	D	10.0.23.2	GigabitEthernet 0/0/2
10.0.23.2/32	Direct	0	0	D	127.0.0.1	GigabitEthernet 0/0/2
10.0.23.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet 0/0/2
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

```
[R2]
```



3.4 Configuring a Backup Static Route

Data exchanged between R2 and the network segments 10.0.13.3 and 10.0.3.3 is transmitted through the link between R2 and R3. If the link between R2 and R3 fails, R2 fails to communicate with the network segments 10.0.13.3 and 10.0.3.3.

However, as shown in the topology, R2 can still communicate with R3 through R1 when the link between R2 and R3 fails. To achieve this, you can configure a backup static route to implement route redundancy. In normal cases, the backup static route does not take effect. It is used only if the link between R2 and R3 fails.

When configuring a backup static route, change the preference value of the backup static route to ensure that it is used only if the primary link fails. In this task, change the preference value of the backup static route to 80.

```
[R1]ip route-static 10.0.3.0 24 10.0.13.3

[R2]ip route-static 10.0.13.0 255.255.255.0 10.0.12.1 preference 80
[R2]ip route-static 10.0.3.0 24 10.0.12.1 preference 80

[R3]ip route-static 10.0.12.0 24 10.0.13.1
```

3.5 Verifying Static Routes

Check the current static route configuration in the routing table of R2.

```
[R2]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
          Destinations : 15          Routes : 15

Destination/Mask    Proto    Pre  Cost           Flags NextHop         Interface
-----
10.0.2.0/24        Direct   0     0             D    10.0.2.2           LoopBack0
10.0.2.2/32        Direct   0     0             D    127.0.0.1          LoopBack0
10.0.2.255/32      Direct   0     0             D    127.0.0.1          LoopBack0
10.0.3.0/24        Static   60     0             RD    10.0.23.3          GigabitEthernet 0/0/2
10.0.12.0/24       Direct   0     0             D    10.0.12.2          GigabitEthernet 0/0/1
10.0.12.2/32       Direct   0     0             D    127.0.0.1          GigabitEthernet 0/0/1
10.0.12.255/32     Direct   0     0             D    127.0.0.1          GigabitEthernet 0/0/1
10.0.13.0/24       Static   60     0             RD    10.0.23.3          GigabitEthernet 0/0/2
10.0.23.0/24       Direct   0     0             D    10.0.23.2          GigabitEthernet 0/0/2
10.0.23.2/32       Direct   0     0             D    127.0.0.1          GigabitEthernet 0/0/2
10.0.23.255/32     Direct   0     0             D    127.0.0.1          GigabitEthernet 0/0/2
127.0.0.0/8        Direct   0     0             D    127.0.0.1          InLoopBack0
127.0.0.1/32       Direct   0     0             D    127.0.0.1          InLoopBack0
127.255.255.255/32 Direct   0     0             D    127.0.0.1          InLoopBack0
255.255.255.255/32 Direct   0     0             D    127.0.0.1          InLoopBack0

[R2]
```



The routing table contains two static routes. The value of the **Protocol** field is **Static**, indicating that the route is a static route. The value of the **Preference** field is **60**, indicating that the default preference is used.

When the link between R2 and R3 is normal, data exchanged between R2 and 10.0.13.3 and between R2 and 10.0.3.3 is transmitted through the link between R2 and R3. Run the **tracert** command to check the data transmission path.

```
<R2>tracert 10.0.13.3
  traceroute to 10.0.13.3(10.0.13.3), max hops: 30 ,packet length: 40,
  press CTRL_C to break
  1 10.0.23.3 40 ms 31 ms 30 ms
<R2>
<R2>
<R2>tracert 10.0.3.3
  traceroute to 10.0.3.3(10.0.3.3), max hops: 30 ,packet length: 40,
  press CTRL_C to break
  1 10.0.23.3 40 ms 30 ms 30 ms
<R2>
```

The command output indicates that R2 directly sends data to R3, bypassing other devices.

3.6 Verifying Backup Static Routes

Disable G0/0/2 on R2 to simulate a link failure between R2 and R3, and then check the changes in the IP routing table.

```
[R2]interface GigabitEthernet 0/0/2
[R2-GigabitEthernet0/0/2]shutdown
[R2-GigabitEthernet0/0/2]quit
[R2]
```

Compare the routing table before and after the interface is disabled.

```
[R2]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
      Destinations : 12          Routes : 12

Destination/Mask    Proto    Pre  Cost           Flags NextHop         Interface
-----
10.0.2.0/24         Direct   0    0              D    10.0.2.2           LoopBack0
10.0.2.2/32         Direct   0    0              D    127.0.0.1          LoopBack0
10.0.2.255/32       Direct   0    0              D    127.0.0.1          LoopBack0
10.0.3.0/24        Static   80    0              RD   10.0.12.1          GigabitEthernet 0/0/1
10.0.12.0/24        Direct   0    0              D    10.0.12.2          GigabitEthernet 0/0/1
10.0.12.2/32        Direct   0    0              D    127.0.0.1          GigabitEthernet 0/0/1
10.0.12.255/32      Direct   0    0              D    127.0.0.1          GigabitEthernet 0/0/1
10.0.13.0/24       Static   80    0              RD   10.0.12.1          GigabitEthernet 0/0/1
127.0.0.0/8         Direct   0    0              D    127.0.0.1          InLoopBack0
127.0.0.1/32        Direct   0    0              D    127.0.0.1          InLoopBack0
127.255.255.255/32  Direct   0    0              D    127.0.0.1          InLoopBack0
255.255.255.255/32  Direct   0    0              D    127.0.0.1          InLoopBack0
```



[R2]

In the routing table of R2, the next hops and preferences of the two routes marked in gray have changed. Check the connectivity from R2 to 10.0.13.3 and 10.0.3.3 on R3.

```
<R2>ping 10.0.3.3
PING 10.0.3.3: 56 data bytes, press CTRL_C to break
Reply from 10.0.3.3: bytes=56 Sequence=1 ttl=255 time=3 ms
Reply from 10.0.3.3: bytes=56 Sequence=2 ttl=255 time=2 ms
Reply from 10.0.3.3: bytes=56 Sequence=3 ttl=255 time=2 ms
Reply from 10.0.3.3: bytes=56 Sequence=4 ttl=255 time=2 ms
Reply from 10.0.3.3: bytes=56 Sequence=5 ttl=255 time=2 ms
--- 10.0.3.3 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 2/2/3 ms

<R2>ping 10.0.13.3
PING 10.0.13.3: 56 data bytes, press CTRL_C to break
Reply from 10.0.13.3: bytes=56 Sequence=1 ttl=255 time=3 ms
Reply from 10.0.13.3: bytes=56 Sequence=2 ttl=255 time=2 ms
Reply from 10.0.13.3: bytes=56 Sequence=3 ttl=255 time=2 ms
Reply from 10.0.13.3: bytes=56 Sequence=4 ttl=255 time=2 ms
Reply from 10.0.13.3: bytes=56 Sequence=5 ttl=255 time=2 ms
--- 10.0.13.3 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 2/2/3 ms

<R2>
```

The network is not interrupted when the link between R2 and R3 is shut down. Run the **tracert** command to check the data packet forwarding path.

```
<R2>tracert 10.0.13.3
tracert to 10.0.13.3(10.0.13.3), max hops: 30 ,packet length: 40,press CTRL_C to break
 1 10.0.12.1 40 ms 21 ms 21 ms
 2 10.0.13.3 30 ms 21 ms 21 ms

<R2>
<R2>tracert 10.0.3.3
tracert to 10.0.3.3(10.0.3.3), max hops: 30 ,packet length: 40,press CTRL_C to break
 1 10.0.12.1 40 ms 21 ms 21 ms
 2 10.0.13.3 30 ms 21 ms 21 ms

<R2>
```

The command output indicates that the data sent by R2 reaches R3 through R1.

3.7 Configuring a Default Route to Implement Network Connectivity

Enable the interface on R2 that was shut down in Step 6.

```
[R2]interface GigabitEthernet 0/0/2
[R2-GigabitEthernet0/0/2]undo shutdown
```



```
[R2-GigabitEthernet0/0/2]quit
[R2]
```

Verify connectivity from R1 to the network segment 10.0.23.3.

```
[R1]ping 10.0.23.3
  PING 10.0.23.3: 56 data bytes, press CTRL_C to break
    Request time out
    Request time out
    Request time out
    Request time out
    Request time out
  --- 10.0.23.3 ping statistics ---
    5 packet(s) transmitted
    0 packet(s) received
  100.00% packet loss
```

R1 does not have a route to the network segment 10.0.23.0. Therefore, packets cannot reach R3.

```
<R1>display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
  Destinations : 14          Routes : 14

Destination/Mask    Proto    Pre  Cost           Flags NextHop         Interface
-----
10.0.1.0/24         Direct   0    0                D    10.0.1.1           LoopBack0
10.0.1.1/32         Direct   0    0                D    127.0.0.1          LoopBack0
10.0.1.255/32       Direct   0    0                D    127.0.0.1          LoopBack0
10.0.3.0/24         Static   60    0               RD    10.0.13.3          GigabitEthernet 0/0/0
10.0.12.0/24        Direct   0    0                D    10.0.12.1          GigabitEthernet 0/0/1
10.0.12.1/32        Direct   0    0                D    127.0.0.1          GigabitEthernet 0/0/1
10.0.12.255/32      Direct   0    0                D    127.0.0.1          GigabitEthernet 0/0/1
10.0.13.0/24        Direct   0    0                D    10.0.13.1          GigabitEthernet 0/0/0
10.0.13.1/32        Direct   0    0                D    127.0.0.1          GigabitEthernet 0/0/0
10.0.13.255/32      Direct   0    0                D    127.0.0.1          GigabitEthernet 0/0/0
127.0.0.0/8         Direct   0    0                D    127.0.0.1          InLoopBack0
127.0.0.1/32        Direct   0    0                D    127.0.0.1          InLoopBack0
127.255.255.255/32  Direct   0    0                D    127.0.0.1          InLoopBack0
255.255.255.255/32  Direct   0    0                D    127.0.0.1          InLoopBack0

<R1>
```

You can configure a default route with the next hop address of 10.0.13.3 on R1 to ensure network connectivity.

```
[R1]ip route-static 0.0.0.0 0.0.0.0 10.0.13.3
```

After the configuration is complete, check the connectivity between R1 and 10.0.23.3.

```
<R1>ping 10.0.23.3
  PING 10.0.23.3: 56 data bytes, press CTRL_C to break
    Reply from 10.0.23.3: bytes=56 Sequence=1 ttl=255 time=3 ms
    Reply from 10.0.23.3: bytes=56 Sequence=2 ttl=255 time=2 ms
```



```
Reply from 10.0.23.3: bytes=56 Sequence=3 ttl=255 time=2 ms
Reply from 10.0.23.3: bytes=56 Sequence=4 ttl=255 time=2 ms
Reply from 10.0.23.3: bytes=56 Sequence=5 ttl=255 time=2 ms
--- 10.0.23.3 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
round-trip min/avg/max = 2/2/3 ms
```

R1 can communicate with the network segment 10.0.23.0 through the default route.

3.8 Configuring a Backup Default Route

If the link between R1 and R3 fails, R1 can use the backup default route to communicate with 10.0.23.3 and 10.0.3.3 through R2. Configure two backup routes to ensure that there are routes for both the forward and return traffic.

```
[R1]ip route-static 0.0.0.0 0.0.0.0 10.0.12.2 preference 80
```

```
[R3]ip route-static 10.0.12.0 24 10.0.23.2 preference 80
```

3.9 Verifying Backup Default Routes

Check the routing entries on R1 when the link is normal.

```
[R1]display ip routing-table
```

Route Flags: R - relay, D - download to fib

Routing Tables: Public

Destinations : 15

Routes : 15

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
0.0.0.0/0	Static	60	0	RD	10.0.13.3	GigabitEthernet 0/0/0
10.0.1.0/24	Direct	0	0	D	10.0.1.1	LoopBack0
10.0.1.1/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.1.255/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.3.0/24	Static	60	0	RD	10.0.13.3	GigabitEthernet 0/0/0
10.0.12.0/24	Direct	0	0	D	10.0.12.1	GigabitEthernet 0/0/1
10.0.12.1/32	Direct	0	0	D	127.0.0.1	GigabitEthernet 0/0/1
10.0.12.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet 0/0/1
10.0.13.0/24	Direct	0	0	D	10.0.13.1	GigabitEthernet0/0/0
10.0.13.1/32	Direct	0	0	D	127.0.0.1	GigabitEthernet 0/0/0
10.0.13.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet 0/0/0
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

```
[R1]
```

Disable the G0/0/0 interfaces on R1 and R3 to simulate a link fault, and then check the routing table of R1. Compare the routing table before and after the interface is disabled.



```
[R1]interface GigabitEthernet0/0/0
[R1-GigabitEthernet0/0/0]shutdown
[R1-GigabitEthernet0/0/0]quit

[R3]interface GigabitEthernet0/0/0
[R3-GigabitEthernet0/0/0]shutdown
[R3-GigabitEthernet0/0/0]quit

[R1]display ip routing-table
Route Flags: R - relay, D - download to fib
```

Routing Tables: Public

Destinations : 11

Routes : 11

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
0.0.0.0/0	Static	80	0	RD	10.0.12.2	GigabitEthernet 0/0/1
10.0.1.0/24	Direct	0	0	D	10.0.1.1	LoopBack0
10.0.1.1/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.1.255/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.12.0/24	Direct	0	0	D	10.0.12.1	GigabitEthernet 0/0/1
10.0.12.1/32	Direct	0	0	D	127.0.0.1	GigabitEthernet 0/0/1
10.0.12.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet 0/0/1
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

[R1]

In the preceding routing table, the preference value of the default route 0.0.0.0 is 80, indicating that the backup default route has taken effect.

```
<R1>ping 10.0.23.3
PING 10.0.23.3: 56 data bytes, press CTRL_C to break
Reply from 10.0.23.3: bytes=56 Sequence=1 ttl=254 time=76 ms
Reply from 10.0.23.3: bytes=56 Sequence=2 ttl=254 time=250 ms
Reply from 10.0.23.3: bytes=56 Sequence=3 ttl=254 time=76 ms
Reply from 10.0.23.3: bytes=56 Sequence=4 ttl=254 time=76 ms
Reply from 10.0.23.3: bytes=56 Sequence=5 ttl=254 time=76 ms
--- 10.0.23.3 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 76/110/250 ms
<R1>
```

The network is not interrupted when the link between R1 and R3 is shut down. Run the **tracert** command to check the data packet forwarding path.

```
<R1>tracert 10.0.23.3
traceroute to 10.0.23.3(10.0.23.2), max hops: 30 ,packet length: 40,press CTRL_C to break
1 10.0.12.2 30 ms 26 ms 26 ms
```



```
2 10.0.23.3 60 ms 53 ms 56 ms
```

The command output shows that the packets are forwarded to R3 (10.0.23.3) through R2 (10.0.12.2).

4 Verification

Verification has been performed during the lab.

5 Quiz

1. How do you configure a static route as a floating static route?
When configuring static routes, change the preference of one static route to configure it as a floating static route.
2. What is the destination network address of a default route?
When configuring a default route, the destination network address is 0.0.0.0, which represents any network.
3. How does a router select the optimal route?
The router first compares preferences of routes. If the routes have the same preferences, the router compares their metrics. If the routes have the same metric, they are added to the routing table as equal-cost routes.